Deep reinforcement learning-guided active control of turbulent flows

Feng Ren [1,2,3] Yuanpu Zhao,[1] Jian Song [1] Boo Cheong Khoo [3] Yongdong Cui [3] Zhaokun Wang [4,5] and Dong Song [1,*]

[1]*School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an, China*
[2]*Shenzhen Research Institute of Northwestern Polytechnical University, Shenzhen, China*
[3]*Department of Mechanical Engineering, National University of Singapore, Singapore*
[4]*Department of Mechanical Engineering, The Hong Kong Polytechnic University, Hong Kong, China*
[5]*School of Fashion and Textiles, The Hong Kong Polytechnic University, Hong Kong, China*

(Received 31 August 2025; accepted 16 March 2026; published 8 April 2026)

In this study, we present a deep reinforcement learning (DRL) framework for closed-loop control of canonical flow past a circular cylinder under fully developed turbulent conditions. To enable computationally efficient training for each episode, we employ a graphics processing unit–accelerated flow solver based on the generalized interpolation-supplemented cascaded lattice Boltzmann method. Moreover, we further employ a two-stage exploration process, beginning with a coarse mesh and minimal spanwise flow resolution before progressing to a finer mesh with well-resolved spanwise flow. This approach significantly reduces computational time, making the implementation of intricate and time-consuming DRL-guided active flow control (AFC) feasible in turbulent flows. The resulting DRL-guided AFC achieves an optimal drag reduction of 55% and a lift fluctuation reduction of 26% due to recovered pressure at the rear side, attenuated wake vortical structures, reduced recirculation bubble length, and mitigated downstream turbulent fluctuations. Additional numerical experiments applying second-order smoothing to the DRL-determined control actions confirm that effective turbulent flow control does not necessarily require high-frequency actuation. This work verifies DRL as a powerful machine-learning tool for tackling challenges of AFC in turbulent flows. The revealed phenomena also highlight rich opportunities for further explorations of the underlying physics of flow under control.

DOI: 10.1103/y7cc-2xh7

I. INTRODUCTION

Active flow control (AFC) represents a long-standing research focus of the fluid mechanics community. By employing actuators to strategically alter flow behavior, AFC aims to enhance aerodynamic or hydrodynamic performance, achieving objectives such as lift augmentation, drag reduction, suppression of flow-induced vibrations, and improvement of mixing or thermal convection [1–4]. AFC strategies are broadly categorized as open-loop or closed-loop, distinguished by whether measured flow information is utilized to dynamically adjust the control action [5]. Compared to open-loop methods, well-designed closed-loop controls offer superior adaptability and effectiveness across a wider range of flow conditions [1,3]. However, designing explicit closed-loop control laws becomes particularly challenging for turbulent flows, characterized by strong nonlinearity and complex interactions across multiple spatial and temporal scales.

*Contact author: songdong1226@nwpu.edu.cn

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Flow past a circular cylinder is a canonical problem in fluid mechanics, yet prevalent in engineering applications such as underwater vehicle appendages, rotary sails, and aircraft landing gears. Its relatively simple geometry and widespread relevance also make it a fundamental model for demonstrating AFC's efficacy [1–4]. Consequently, AFC applications targeting drag reduction or lift fluctuation suppression for cylinder flows have garnered significant interest and recognition as effective approaches in recent years [1,2]. This flow exhibits complex phenomena, including separation, wake transition, and boundary layer transition. Investigating closed-loop control for this well-defined geometry provides critical insights applicable to flow control around other bluff bodies [1]. While foundational studies exist, realizing effective closed-loop control remains challenging, primarily due to difficulties in accurately characterizing complex flow states from limited sensor feedback, especially within the fully turbulent regime [1,3].

Proportional-integral-derivative (PID) control remains a widely implemented closed-loop strategy due to its simplicity [3,6]. Nevertheless, it faces critical limitations in flow control applications. Its linear actuation mechanism, based on feedback signals, exhibits high sensitivity to control parameters, typically requiring laborious experimental tuning or extensive engineering experience. For instance, while velocity information is commonly used as feedback to suppress vortex shedding, optimizing the high-dimensional parameter space (gains, sensor locations, phase delays) is complicated by the strong sensitivity of flow structures to minor parameter changes [7–9]. This makes identifying optimal parameters across varying Reynolds numbers (Re) particularly challenging. Methods such as iterative tuning [10] or sensor-location optimization [11] offer only partial solutions. Furthermore, PID control is inherently ill-suited for multiobjective control. For example, Lu *et al.* [12] used the instantaneous lift coefficient as feedback for cylinder rotation, achieving a 50.7% reduction in lift fluctuations but only a modest 10% drag reduction at a Reynolds number below 200. Song *et al.* [13] conducted experimental investigations of closed-loop rotary control for the wake of a circular cylinder at $\mathrm{Re} = 100$ with proportional control, successfully mitigating the amplitude of monitored transverse velocity by 52%. With proportional control and spanwise-averaged velocity as a feedback signal, Yun and Lee [14] carried out jet control over a circular cylinder at $\mathrm{Re} = 3900$. The mean drag and lift fluctuations are reduced by about 11% and 61%, respectively, with the optimal control parameters. However, given the highly nonlinear nature of turbulent flow, it is questionable whether PID control that is rooted in linear theory can provide a generalized solution for adaptive turbulent flow control.

Deep reinforcement learning (DRL), a rapidly evolving machine learning paradigm, offers significant promise for advanced closed-loop flow control [3,15–20]. Within a DRL framework, an agent learns an optimal control policy through iterative trial-and-error interactions with a simulated or real flow environment. Executing actions based on observed flow states, the agent aims to maximize cumulative rewards. Leveraging deep neural networks, DRL can process high-dimensional flow data and generate complex, nonlinear control laws without requiring explicit flow models. Pioneering work by Rabault *et al.* [21] demonstrated DRL-based closed-loop control for cylinder flow at $\mathrm{Re} = 100$, achieving 8% drag reduction using a proximal policy optimization (PPO) agent [22] driven by velocity feedback. This approach was extended to $\mathrm{Re} = 400$ by Tang *et al.* [23], attaining 38.7% drag reduction with four actuators. Subsequent studies improved training efficiency [24] and optimized sensor placement [25]. However, applications of DRL for performance enhancement in fully turbulent flows remain relatively scarce [26–29]. Ren *et al.* [27] reported around 30% drag reduction numerically for two-dimensional flow at $\mathrm{Re} = 1000$. It is crucial to note that two-dimensional turbulence exhibits distinct phenomena such as inverse energy cascade and is not a simple reduction of three-dimensional flow [30]. Amico *et al.* [28] experimentally applied DRL to a square-back road vehicle ($\mathrm{Re} \sim 10^5$), achieving 10% drag reduction but cautioning that their single-step approach might lack generalization for highly unsteady flows. A multistage training approach within a multiagent reinforcement learning (RL) framework was developed for a three-dimensional cylinder at $\mathrm{Re} = 3900$ [29], yielding approximately 9% drag reduction at low energy cost, but at the expense of increased lift fluctuations. In the fully turbulent regime, Fan *et al.* [26] applied RL to actively control flow around a bluff body in both experiments and simulations.

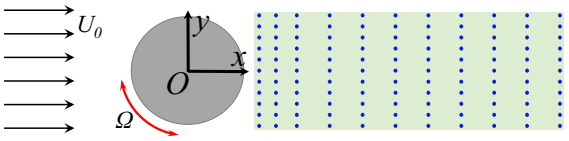


FIG. 1. Schematic of the active flow control configuration: Uniform upstream flow with velocity U_0, fixed cylinder with controllable rotational velocity Ω, and wake sensor array providing velocity feedback.

Their strategy involved rotating two rods near the cylinder's separation point. Using a similar setup, Wang *et al.* [31] employed transfer learning to adapt the RL policy from a lower-Reynolds-number regime to the subcritical regime. While DRL has demonstrated effectiveness for diverse control objectives [32–36], its application and assessment in high-Re, three-dimensional turbulent flows require further exploration.

Regarding the above background, this study aims to showcase the feasibility and effectiveness of a DRL-guided approach for closed-loop control of flow past a circular cylinder under fully developed turbulent conditions. The DRL agent generates a control policy dictating cylinder rotation to *simultaneously* reduce the mean drag force and suppress lift fluctuations.

II. METHODOLOGY

A. Flow configuration

This study employs the canonical flow past a circular cylinder to demonstrate deep reinforcement learning (DRL)–guided active flow control (AFC). As shown in Fig. 1, a cylinder of diameter D_0 is immersed in a uniform incoming flow with velocity U_0. The Reynolds number is defined as $\mathrm{Re} = U_0 D_0/\nu$, where ν denotes the fluid kinematic viscosity, while the Mach number is fixed at $\mathrm{Ma} = U_0/c_s$, with c_s being the speed of sound. Throughout this work, Re and Ma are maintained at 10 000 and 0.088, respectively, establishing fully turbulent and weakly compressible flow conditions. At this Reynolds number, the flow is in the fully turbulent regime. In addition, a lot of prior research related to flow past a circular cylinder was carried out in this condition [37,38], thus allowing validations of the flow solver and a deep understanding of the flow physics before conducting the AFC. At the corresponding Mach number, the maximum local Mach number can be roughly limited to 0.3 (considering tangential velocity at the edge of cylinder surface) so that the flow is assumed to be incompressible. Throughout the whole study, quantities and results are presented in nondimensional form using U_0, D_0, and the reference timescale $T_0 = D_0/U_0$.

An array of 121 velocity sensors (blue dots in Fig. 1) are used to monitor the flow field and provide feedback signals, each providing time-dependent streamwise and transverse velocity components. Spanwise velocity is excluded to replicate typical two-dimensional, two-component (2D2C) particle image velocimetry measurements. The velocity sensors are positioned at dimensionless coordinates, i.e., $x \in 0.6, 0.8, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0$ and $y \in 0.0, \pm 0.1, \pm 0.2, \pm 0.3, \pm 0.4, \pm 0.5$. Here the cylinder's center is set as the original coordinate $\{0, 0\}$. Rotary control is applied to the cylinder with rotational velocity Ω, which is normalized by U_0/D_0 and controlled in a real-time manner by the DRL agent, so as to modulate near-field flow dynamics. Ω is constrained to an interval of $[-6, 6]$, limiting the surface tangential velocity to three times the upstream flow velocity.

The DRL agent aims to simultaneously reduce drag and suppress lift fluctuations. Accordingly, we adopt a reward function consistent with prior studies [21,24,27], i.e.,

$$r = -\langle C_D \rangle_S - w\langle |C_L| \rangle_S, \tag{1}$$

where C_D and C_L denote drag and lift coefficients, respectively. The operator $\langle \cdot \rangle_S$ represents averaging between two actuations. The weighting factor $w = 2$ is used to penalize significant lift biases or fluctuations while encouraging drag reductions when exploring the optimal strategy. As

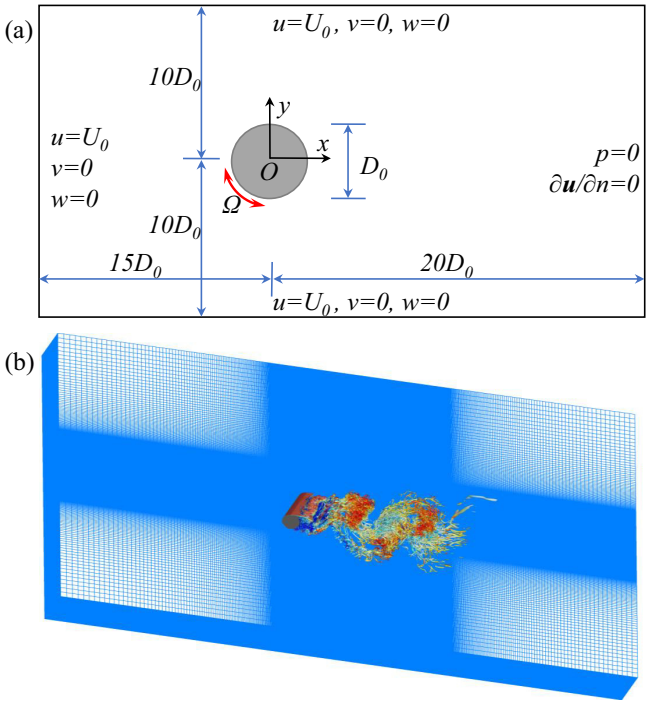


FIG. 2. (a) Schematic of the computational domain and boundary conditions in the Oxy plane. (b) Mesh partition with refined resolution around the cylinder and in the wake.

discussed in prior studies [21,27], a relatively large value of w (e.g., $w > 1$) avoids degenerated control strategies with unwanted lift biases in laminar or weakly turbulent conditions. Since the ratio of mean drag to the root mean square (RMS) of lift is approximately 2 in the uncontrolled case, an empirical weighting factor of $w = 2$ is chosen to maintain a balance between minimizing drag and mitigating lift fluctuations.

B. Flow solver

In this study, we extend the two-dimensional generalized interpolation-supplemented cascaded lattice Boltzmann method (GICLBM) [39,40] to the three-dimensional situation. The GICLBM is a variant of the lattice Boltzmann method (LBM) that combines the cascaded collision operator and the generalized form of interpolation-supplemented streaming process, and it has been verified to be sufficiently reliable for simulations of complex flow problems. In the algorithm, the collision step employs the central-moment-based or cascaded collision operator that incorporates high-order moments and preserves Galilean invariance [41]. This collision operator shows superior numerical stability to that of the conventional single- or multiple-relaxation time collision operator, which is critical in simulating high-Reynolds-number flows [42].

A nonuniform Cartesian mesh balances computational efficiency and accuracy across near- and far-field regions. This is achieved through the generalized interpolation-supplemented streaming step, where the particle streaming process of a conventional LBM that is along a fixed path of a cubic lattice is replaced by a second-order interpolation. Here the computational domain still uses a uniform Cartesian mesh, while the physical domain can use a nonuniform mesh to enable localized mesh refinement in both near-wall and the concerned wake region. Further introductions of the GICLBM can be found from Refs. [43,44]. As shown in Fig. 2(b), the cylinder and near-wake region ($-1.5 \leqslant x \leqslant 8$, $-2.25 \leqslant y \leqslant 2.25$) employ uniform fine meshes. Spanwise (z-direction)

TABLE I. Validation of the flow solver and mesh convergence study.

	Configuration	$D_0/\delta x$	$T_0/\delta t$	L_z/D_0	$\overline{C_D}$	$C_{L,\mathrm{rms}}$	St	$-\overline{C_{pb}}$
	DNS-A3				1.128	0.574	0.205	1.171
Dong and Karniadakis [37]	DNS-B1				1.208	0.547	0.200	1.201
	DNS-B3				1.143	0.448	0.203	1.129
Jiang *et al.* [38]					1.19	0.47	0.197	1.21
	Mesh I	50	1000	0.16	1.214	0.891	0.207	
Present	Mesh II	60	1200	1	1.273	0.603	0.205	
	Mesh III	80	1600	3	1.261	0.560	0.192	1.229

Note: $\delta x = 1$ and $\delta t = 1$ are the lattice unit length and time step, respectively.

meshing remains uniform. In the other regions, the mesh spacing is exponentially coarsened to reduce unnecessary computational cost.

The computational domain and boundary conditions in the Oxy plane are sketched in Fig. 2(a). Herein a uniform velocity profile is applied at the inlet and the lateral boundaries. At the outlet, a zero-pressure condition is applied alongside a zero-gradient condition for the normal velocity. Both BCs are implemented using the nonequilibrium extrapolation scheme [45]. Periodic conditions are applied at the spanwise boundaries.

To model the BC of the cylinder with/without rotary control, we apply the direct-forcing immersed boundary method (IBM) with multiple iterations [46]. In the IBM, the curved wall boundary is modeled with a group of Lagrangian nodes, where the no-slip and no-penetration conditions of these boundary nodes are transformed into body forces exerted on the surrounding Eulerian nodes. The Euler-Lagrange mapping process is iterated five times to reduce numerical errors. The hydrodynamic forces exerted on the cylinder are then computed via integrating the Lagrangian forces.

In terms of turbulent modeling, we resort to large-eddy simulations (LES). The flow discussed here is a typical type of separated flow, a class of problem that is challenging for Reynolds-averaged Navier-Stokes (RANS) models. We argue that for DRL-guided AFCs, it is essential to provide well-resolved flow data whenever the computational cost is acceptable. The inherent approximations of less accurate models could possibly mislead the DRL agent, compromising both the derived optimal strategy and physical interpretations thereafter. In the LES, the Vreman subgrid-scale model [47] is incorporated, where the fluid viscosity involves the molecular viscosity and extra eddy viscosity, the latter modeling subgrid-scale dissipation and being derived from a local velocity derivative tensor. Velocity derivatives are numerically computed using the second-order finite-difference scheme. This turbulent modeling approach has been successfully implemented and verified in our previous works [48,49].

To validate the current flow solver, we conduct simulations using three different configurations and make comparisons with results from prior studies [37,38], as summarized in Table I. Here Mesh I, Mesh II, and Mesh III utilize $512 \times 320 \times 8$, $640 \times 384 \times 60$, and $1024 \times 512 \times 240$ nodes, respectively. In the direct numerical simulation (DNS) results of Dong and Karniadakis [37], the DNS-A3 configuration uses fewer spectral elements than the DNS-B3 configuration, while the DNS-B1 configuration uses fewer grid points in the spanwise direction. By comparing the numerical results, one can note that the mean C_D, C_L fluctuation, the vortex shedding Strouhal number St, and the base pressure coefficient measured at the rear of the cylinder surface are fairly consistent with results from Refs. [37,38]. The median mesh setup (Mesh II) generates $\overline{C_D}$ and $C_{L,\mathrm{rms}}$ values comparable to those obtained with the finest mesh (Mesh III). Mesh I's coarser resolution, as well as very few spanwise mesh nodes that resolve least spanwise flow information, degrades lift fluctuation predictions, though mean drag and St remain acceptable.

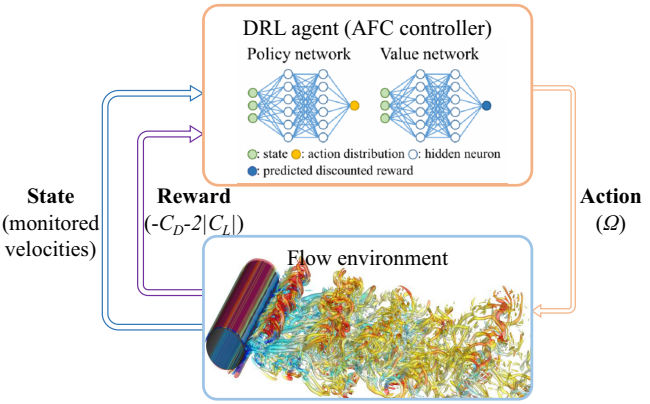


FIG. 3. Schematic of the DRL control loop. The DRL agent (AFC controller) comprises policy and value networks. The flow environment is simulated using our GICLBM-based solver. States, actions, and rewards are collected during the interaction process.

For the DRL training, well balancing the computational accuracy and the computational efficiency is vital. For this aim, Mesh I and Mesh II are adopted during the training stage, where each episode runs for a duration of $100T_0$. Once the training reaches convergence and effective control strategies are identified, the settings in the highly resolved Mesh III are used to evaluate the selected control strategies, when the computational efficiency is no longer a big concern. This study leverages NVIDIA's compute unified device architecture (CUDA) on an RTX 4090 GPU device to accelerate our in-house code. For a representative simulation that lasts $100T_0$, the required computation time is reduced to approximately 5 min (Mesh I), 90 min (Mesh II), and 24 h (Mesh III). For DRL-guided AFC carried out in the numerical environment, computational time and hardware cost are key factors. Compared to the work of Suárez *et al.* [29], which requires training on 90 CPU nodes concurrently for a minimum of 2 weeks, our training demonstrates superior computational efficiency and very low computational cost.

C. Deep reinforcement learning

The closed-loop interaction between the DRL agent and the flow environment is depicted in Fig. 3. A sensor array collects velocity information (i.e., the state) from the simulation, while the cylinder employs rotary actuation (i.e., the action) to modify the flow field. Control performance is evaluated using the reward function defined in Eq. (1). We implement the DRL agent using the PPO algorithm [22].

The DRL agent utilizes two artificial neural networks (ANNs), i.e., a policy network and a value network, which are also known as "actor" network and "critic" network, respectively. In our in-house DRL code, the TensorFlow library [50] is used to establish the networks. The policy network serves as the AFC controller, with states as input and actions as output. The value network predicts long-term performance given the current state as input. Both networks contain two hidden layers, each having 320×320 neurons. When updating the policy, we use the Adam (short for "adaptive moment estimation") optimizer [51]. During the training, actions are sampled from certain probability distributions whose characteristic parameters are determined by the policy network, and the Beta distribution is utilized for this aim [52]. Unlike the more frequently used Gaussian distribution, the Beta distribution is inherently bounded, and this property allows for the direct enforcement of a specified actuation range. Once the training converges and the performance reaches a satisfactory level, deterministic runs are executed, using the mean action from the policy distribution without stochastic sampling. Except for the number of neurons of hidden layers, all

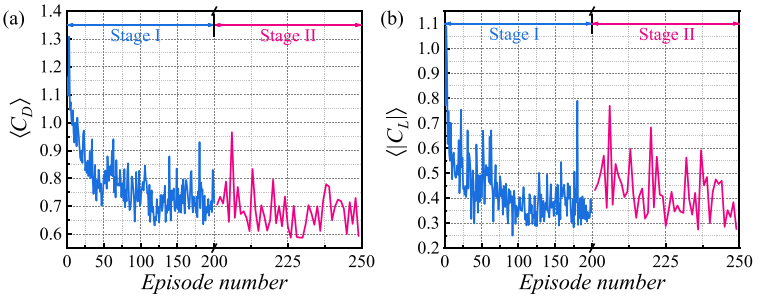


FIG. 4. Variations of (a) mean drag coefficient $\langle C_D \rangle$ and (b) mean lift fluctuation magnitude $\langle |C_L| \rangle$ during the two-stage training.

hyperparameters, including learning rates, the activation function, etc., are kept consistent with those in our prior work [34,35].

III. RESULTS AND DISCUSSIONS

A. Two-stage DRL training

In the DRL, the training process is organized into episodes, each representing an independent trial. At the start of each episode, the flow field is restored from a fully developed, uncontrolled flow and then advances for a randomized duration between 0 and $5T_0$ (approximately a vortex-shedding cycle). This ensures that every episode begins from a random phase within a complete vortex-shedding period [53]. Since DRL training often requires hundreds or even thousands of episodes to achieve satisfactory control performance, the computational time per episode is critical. For two-dimensional cases, our GPU-accelerated flow solver completes an episode within a few minutes [27,33–35], enabling efficient explorations of control strategies. However, achieving successful DRL training under three-dimensional, fully developed turbulent conditions presents significant challenges. This necessitates problem simplification, which we believe should be based on one's physical understanding of the specific problem.

For flow past a circular cylinder, we first assume the spanwise flow is weak relative to the flow inside any 2D plane that consists of mainly the streamwise and the transverse flow. We also assume that the lift and drag are governed primarily by the 2D flow dynamics. This is basically adequate, as in some engineering applications the spanwise flow is neglected. Moreover, in prior studies that rigorously discuss the 3D effect, the 2D flow still retains most features of the 3D flow, such as the vortex shedding and pressure recovery, and the drag and lift forces of a 2D flow are comparable to those from 3D simulations [54,55]. For $\mathrm{Re} = 10^3 - 10^6$, the 2D and 3D simulations of flow past a stationary cylinder predict the drag coefficients and drag crisis very closely [54]. Consequently, the training first employs minimal spanwise resolution (Mesh I, Sec. II B), allowing the DRL agent to rapidly identify control strategies under this simplification. Subsequent simulations with moderate spanwise length ($L_z/D_0 = 1$) then refine these strategies. This rationale motivates our two-stage training approach, which enhances learning efficiency without sacrificing mesh-resolved information:

(i) Stage I: Training uses coarse spatial resolution and a limited spanwise domain (Mesh I) to accelerate convergence to coarse-grained turbulent strategies while neglecting the spanwise effect to a large extent. Each episode requires approximately 5 min.

(ii) Stage II: Training proceeds via restoring strategies from Stage I and employs refined spatial resolution and larger spanwise domain (Mesh II) to resolve more turbulent structures and spanwise flow dynamics. Each episode requires approximately 90 min.

As shown in Fig. 4, the complete training comprises 200 episodes in Stage I and 50 episodes in Stage II, with the latter initialized using strategies from Stage I. Episode counts are adjustable during

training. Despite strong nonlinearity, the training exhibits clear gradual convergence, indicating progressive consolidation of exploration gains. This convergence pattern suggests that later episodes constitute a fine-tuning phase, where successive strategies approach optimal performance. In Stage II, the finer mesh and extended domain resolve more detailed flow structures. In contrast, the coarser mesh in Stage I increases subgrid-scale dissipation, yielding attenuated and less accurate data. This, in turn, leads to the more volatile converging trend of Stage II than that observed during training in Stage I. From the whole trend, Stage II indeed improves the control performance, although this improvement is less significant. Ultimately, the two-stage DRL achieves roughly 50% drag reduction and a significant lift mitigation, and the exact performance would be confirmed through deterministic tests in the next section.

The entire training consumed approximately 90 h on a single RTX 4090 GPU, representing significant computational savings in both time and hardware cost. Further reductions can be achieved through coarser meshes or fewer episodes, especially in Stage II. Thus, our two-stage strategy confirms efficient DRL training that is feasible for 3D fully turbulent flows past a circular cylinder and similar problems in the future.

We subsequently validate control performance via systematic comparisons on a finer mesh (Mesh III). After the training is completed, deterministic control tests are conducted using this high-accuracy configuration. To demonstrate the necessity of two-stage training and evaluate the obtained strategies in detail, two representatives from the converging learning trends are selected, i.e., the 193rd episode from Stage I and the 249th episode from Stage II, due to their marginally superior performance in drag reduction and lift mitigation compared to adjacent episodes. Note that in DRL-guided active flow control, selecting a final policy from the converged training curve remains an empirical step. While strategies from different late-stage episodes should theoretically show marginal differences, they can still vary in practice, particularly in fully developed turbulent conditions. Here we choose an episode where the control performance is among the best alternatives and the performance is robust. For the later tip, the neighboring episodes also achieve a similar performance level, indicating that the strategy does not vary much with different initial conditions. The selected representative episodes are hereafter denoted as Ep 193 and Ep 249, respectively. In addition, a supplementary training trial that adopts identical setups to those in Fig. 4 is presented in the Appendix, so as to reveal the overall similar DRL exploration trend.

B. Deterministic control

Instantaneous flow field snapshots are presented in Fig. 5, with columns corresponding to the uncontrolled case, and DRL-controlled cases at the 193rd and 249th training episodes, respectively. These snapshots are captured at instants of local lift force maxima. Analysis of 3D vortical structures and spanwise vorticity in the central Oxy plane reveals significant flow modifications in controlled cases, including attenuated vortical structure strength and reduced coherent structure size. In addition, the shear layers that appear in the vicinity of the cylinder's upper and lower surface in the uncontrolled cases are nearly eliminated in both DRL-controlled cases. These observations suggest weakened cylinder-vortex interactions and potentially elevated vortex shedding frequencies.

Pressure recovery is also clearly evident on the rear side of the cylinder, as depicted in Figs. 5(b2) and 5(c2), with the 249th DRL-guided strategy demonstrating superior performance. Crucially, the low-pressure region on the rear side is the primary source of drag. Enhanced pressure recovery thus provides clear evidence that drag is significantly reduced in the controlled case.

A reduction in the recirculation bubble size is observable in both the streamwise velocity contours and the vortical structure's color of Figs. 5(a1)–5(c1). This reduction probably stems from momentum transported laterally by the cylinder's alternating rotation to the rear side, compensating the momentum deficit within the original recirculation region. Notably, this bubble size reduction contrasts with observations from our prior DRL-guided AFC studies [27,35], where an elongated recirculation bubble was associated with drag reduction and mitigated lift. The distinct phenomena with similar control performance that will be revealed later indicate rich physics behind this complex

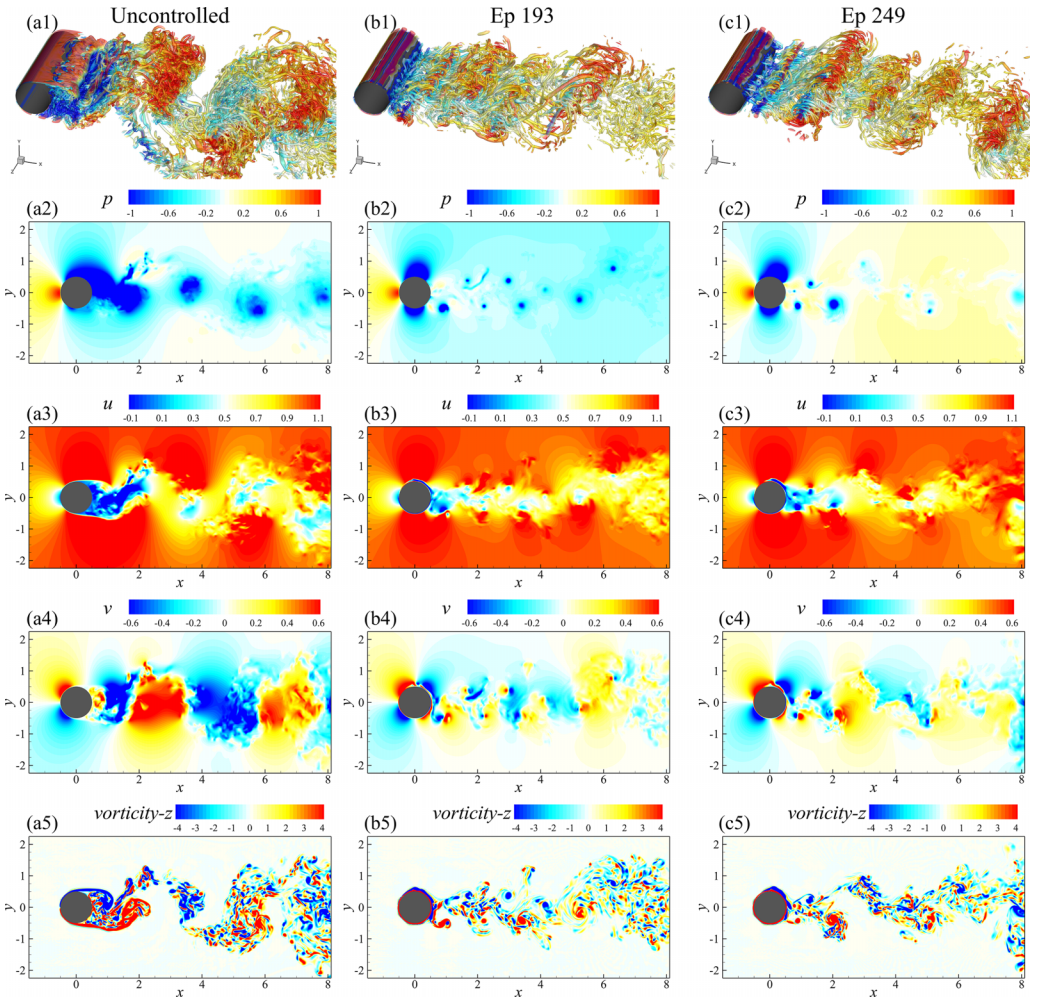


FIG. 5. Instantaneous snapshots of flow fields of the uncontrolled case and the DRL-controlled case learned at the 193rd and 249th episode. (a1)–(c1) Three-dimensional vortical structures identified by the Q criterion and colored by the streamwise velocity u. (a2)–(c2) Pressure p distributed in the Oxy plane at $z = 0$. (a3)–(c3) Streamwise velocity u. (a4)–(c4) Transverse velocity v. (a5)–(c5) The zth component of vorticity.

flow system and require thorough extrapolations in the future. Furthermore, attenuated transverse velocity in Figs. 5(a4)–5(c4) reflects substantially reduced momentum transport within the wake, a key factor directly linked to the mitigated lift fluctuations acting on the cylinder.

In contrast to a prior study of Suárez *et al.* [29] where the control actions are applied at specified positions along the spanwise direction, our method applies a uniform rotational velocity along the entire cylinder. Figure 6 illustrates instantaneous isosurfaces of streamwise velocity, which is commonly used to visualize the recirculation bubble. For the uncontrolled case, the spanwise unevenness is evident, due to the 3D effect that occurs at Reynolds numbers higher than 180. Under control, however, differences along the spanwise direction observed from the shape and color of the recirculation bubble become marginal, indicating that the DRL-guided strategies synchronize the near-field flow evolution across different spanwise locations.

Figure 7 illustrates time histories of the drag coefficient C_D and lift coefficient C_L, along with their corresponding probability density functions (PDFs). Simulations span 300 reference time units,

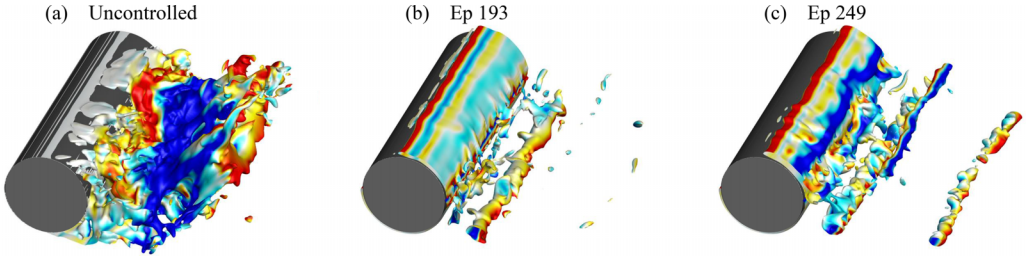


FIG. 6. Instantaneous isosurfaces of streamwise velocity ($u = 0$) colored by the transverse velocity: (a) the uncontrolled case, and the DRL-controlled case learned at the 193rd (b) and 249th episode (c). The normalized transverse velocity ranges from -0.6 to 0.6.

initialized with fully developed uncontrolled flow fields. Both AFC strategies are turned on at $t = 0$, reaching equilibrium by $t \approx 50$. Analysis of time-varying data from $t = 50$ to 300 shows mean drag reductions of 37% and 55% for the 193rd and 249th episodes, respectively. Note that in the work of Wang *et al.* [31] where the Reynolds number is at 10^4, the minimum mean drag is slightly lower than 0.7, while the value is 0.57 in the present study, indicating an improved control performance. The PDF of the drag coefficient indicates augmented fluctuations. Quantified by the RMS of C_D, these fluctuations increase by 168% and 47% for the 193rd and 249th episodes, respectively. Because the RMS of C_D in the uncontrolled baseline is inherently low, particularly when compared to other quantities, its minimization is not typically a focus. While the control method does amplify this fluctuation, this is evaluated as a secondary effect. Lift fluctuations, however, play a more important role in many realistic applications and are a critical factor in our AFCs. Quantified by the RMS of C_L, the lift fluctuations decrease by 26% for the 249th episode, while they increase slightly by 14% for the 193rd episode. These results confirm the efficacy of DRL-guided AFC. Moreover, the results also confirm the necessity of Stage II in the training.

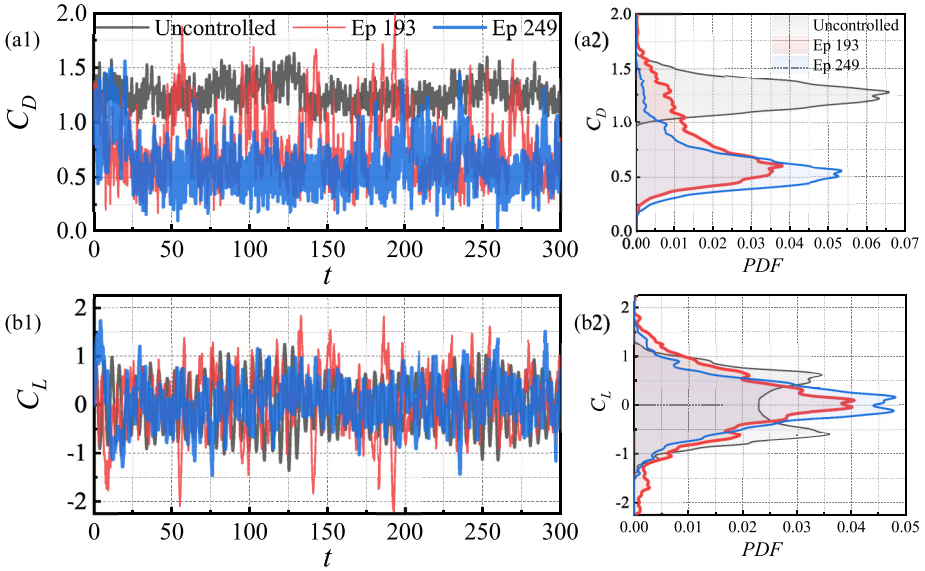


FIG. 7. (a) Time histories of drag coefficient C_D and lift coefficient C_L. (b) PDF of C_D and C_L.

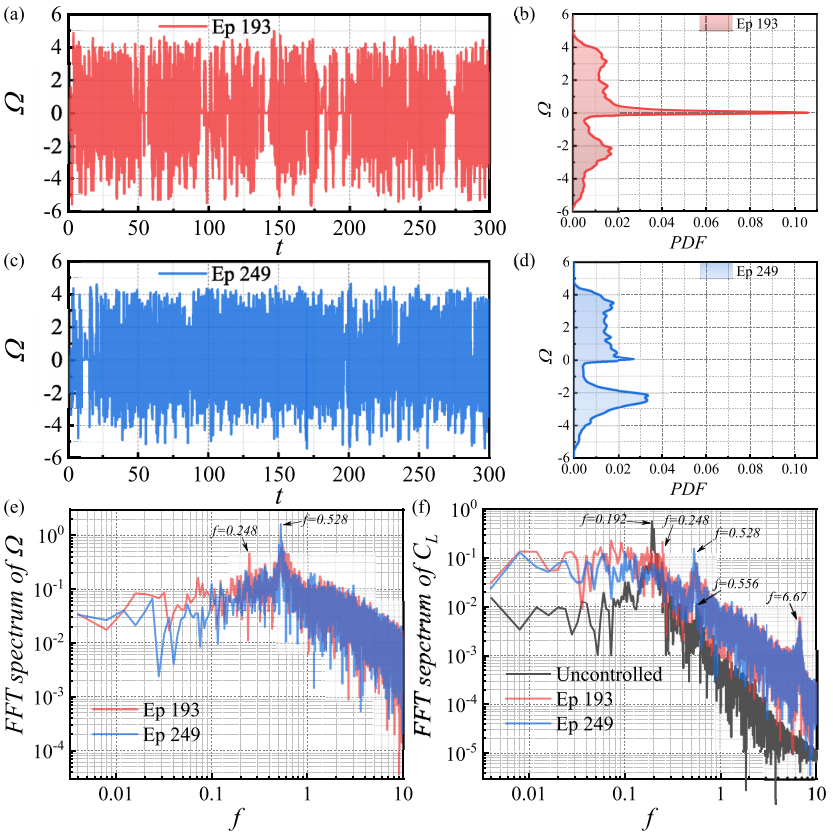


FIG. 8. (a) Time histories of rotational velocity Ω of the 193rd episode. (b) PDF of Ω of the 193rd episode. (c) Time histories of rotational velocity Ω of the 249th episode. (d) PDF of Ω of the 249th episode. (e) FFT spectrum of Ω. (f) FFT spectrum of C_L for reference.

Time histories of the rotational velocity Ω, along with their PDF and FFT results, are presented in Fig. 8. For comparative purposes, the lift coefficient spectrum is included in Fig. 8(d). The Ω signals exhibit strong fluctuations in the time domain with broadband spectral characteristics. Notably, the control strategy from the 193rd episode generates Ω values that intermittently approach zero, indicated by sharp crests in Fig. 8(b). From the C_L spectrum of the uncontrolled case, the vortex shedding frequency $f = 0.192$ plays a dominating role, while a secondary frequency $f = 0.556$ is also observed, which is clearly not a harmonic of the vortex shedding frequency. We deduce that the frequency is related to the characteristic frequency of shear layer instabilities, commonly known as Kelvin-Helmholtz (KH) instability. According to the power law $f_{\mathrm{KH}}/f_{\mathrm{vs}} = 0.0235\,\mathrm{Re}^{0.67}$ proposed by Prasad and Williamson [56] (where f_{vs} denotes the vortex shedding frequency and f_{KH} the KH instability frequency), the predicted KH frequency for the uncontrolled case is 2.16, being much higher than the vortex shedding frequency. Direct evidence to link this frequency specifically to the KH instability would require a comprehensive stability analysis that simultaneously accounts for vortex shedding, shear layer instability, and even the spanwise instability modes, which would be investigated in our future work.

In the controlled cases, both DRL control strategies show a dominant spectral peak at $f = 0.528$ in Fig. 8(b). While a secondary spectral peak appears in the 193rd episode, it is considered less significant as it vanishes in the 249th episode. The frequency $f = 0.528$ differs from the uncontrolled vortex shedding frequency but closely aligns with the aforementioned secondary frequency

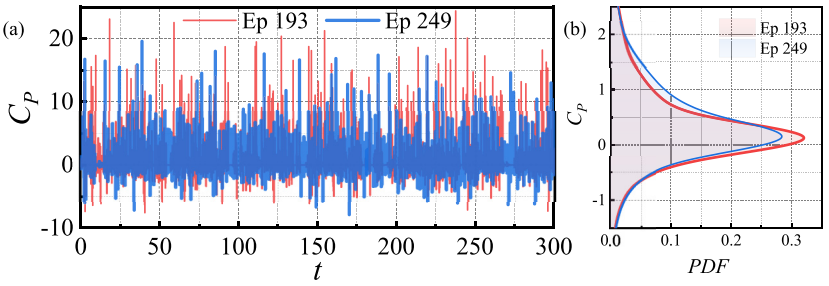


FIG. 9. (a) Time histories of power coefficient C_P of the DRL-controlled cases from the 193rd episode and the 249th episode. (b) PDF of C_P.

$f = 0.556$. Recalling the instantaneous vortical structures in Figs. 5(a1)–5(c1) and Figs. 5(a5)–5(c5), it is revealed that the uncontrolled shear layers forming near the cylinder surface become disrupted under DRL-guided AFC, which we believe is due to the control frequency observed herein. Through the above analyses, it is demonstrated that the DRL agent identifies and leverages a subtle frequency to achieve its control objectives.

To quantify energy consumption in the AFC, we define the power coefficient as $C_P = -\Omega C_M$, where C_M denotes the torque exerted on the cylinder during rotary actuation. The negative sign ensures that positive C_P values indicate energy input required to drive rotation, while negative values correspond to energy harnessing. Figure 9 presents time histories of C_P along with the associated PDF. Negative C_P indicates that the DRL-guided control frequently leverages torque that aligns with the rotational direction, resulting in negative C_P values that require no external power input. This demonstrates the energy-efficient nature of the DRL approach. Despite substantial temporal fluctuations, the mean C_P values remain low, specifically 0.44 and 0.49 for Episode 193 and Episode 249, respectively. Note that C_P is an *a posteriori* quantity derived from controlled flow, thus being difficult to evaluate before performing the controlled cases. However, it establishes a baseline that would allow for quantitative comparisons in future studies that employ similar actuation methods.

First- and second-order turbulent statistics in the Oxy plane are presented in Fig. 10, with statistical averaging performed over $t = 50$–300. The pressure recovery previously observed in Figs. 5(a2)–5(c2) is confirmed again by time-averaged pressure fields in Figs. 10(a1)–10(c1), where the rear-side pressure for the 249th strategy substantially exceeds that of the 193rd strategy. Augmented pressure fluctuations near the cylinder surface [Figs. 10(a4)–10(c4)] primarily stem from highly fluctuating rotational velocities. Notably, intense pressure fluctuations are only predominant in the vicinity of the lateral surfaces, particularly for the 249th-episode strategy, while downstream regions exhibit attenuated pressure fluctuations. This demonstrates that DRL-guided AFC effectively suppresses turbulent fluctuations beyond the actuator region.

Similar fluctuation suppression phenomena appear in other second-order turbulent statistics. Contours of $\overline{u'u'}$, $\overline{v'v'}$, and $\overline{-u'v'}$ reveal enhanced fluctuations near the cylinder surface, while significantly attenuated fluctuations appear in downstream regions. The time-averaged streamwise velocity demonstrates that rotary control compensates momentum near the rear side of the cylinder, significantly reducing the recirculation bubble size. Additionally, the wake width, which is measured vertically between the $u = 1$ contour lines, narrows under control. These findings collectively demonstrate that DRL-guided AFC simultaneously reduces drag, mitigates lift fluctuation, and suppresses turbulent intensity in the wake. As wake turbulence suppression is highly desirable in some practical applications, this study offers a viable strategy.

While turbulent statistics confirm suppression effectiveness, their spectral validity requires further verification. Figure 11 therefore presents turbulent kinetic energy (TKE) spectra at two representative points, i.e., (1, 0, 0) and (4, 0, 0), which are located, in the uncontrolled cases, within and beyond the recirculation bubble, respectively. For frequencies $f > 0.3$, both controlled cases

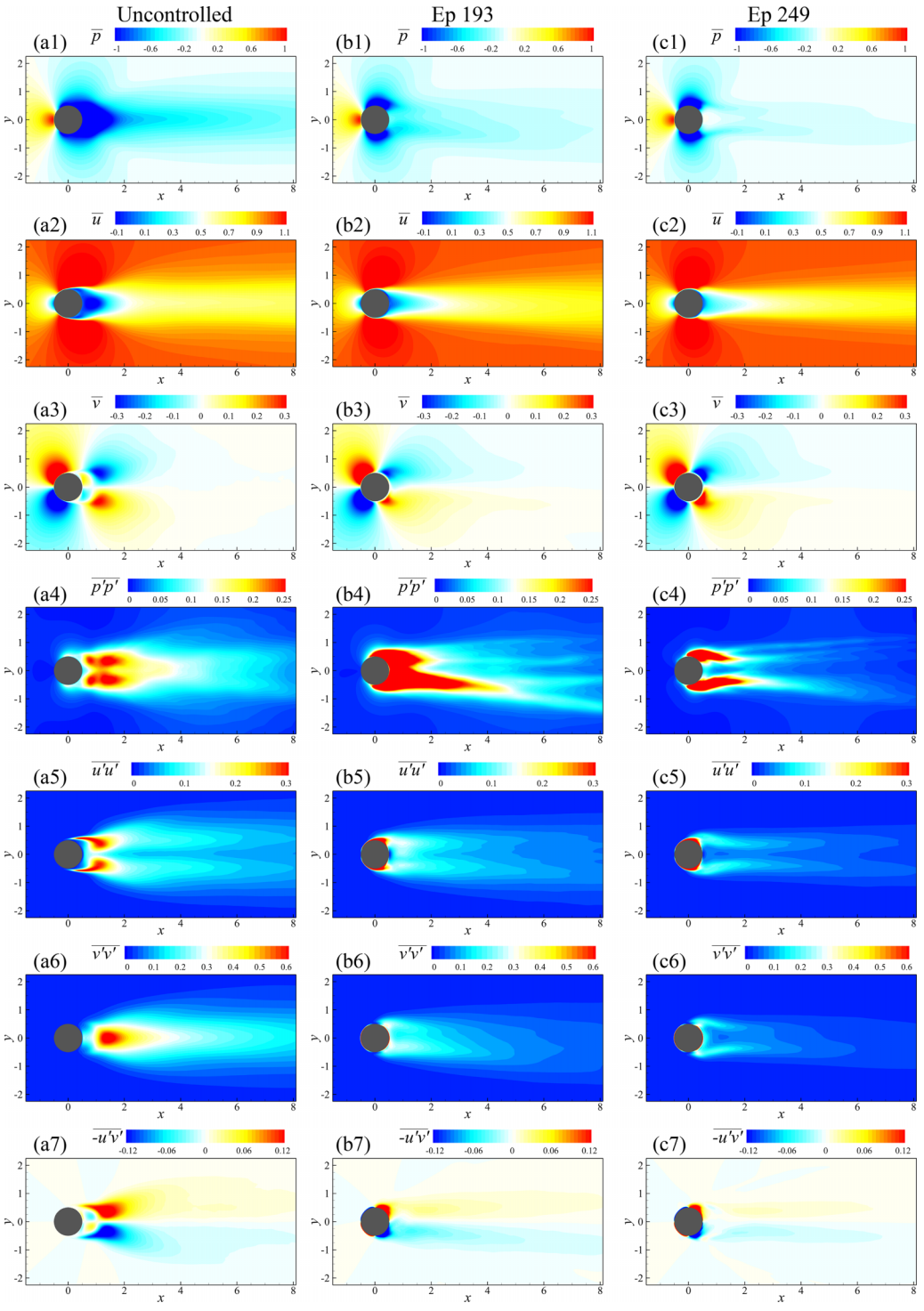


FIG. 10. Turbulent statistics of the uncontrolled case (left), the 193rd episode (middle), and the 249th episode (right). (a1)–(c1) $\overline{p}$. (a2)–(c2) $\overline{u}$. (a3)–(c3) $\overline{v}$. (a4)–(c4) $\overline{p'p'}$. (a5)–(c5) $\overline{u'u'}$. (a6)–(c6) $\overline{v'v'}$. (a7)–(c7) $-\overline{u'v'}$.

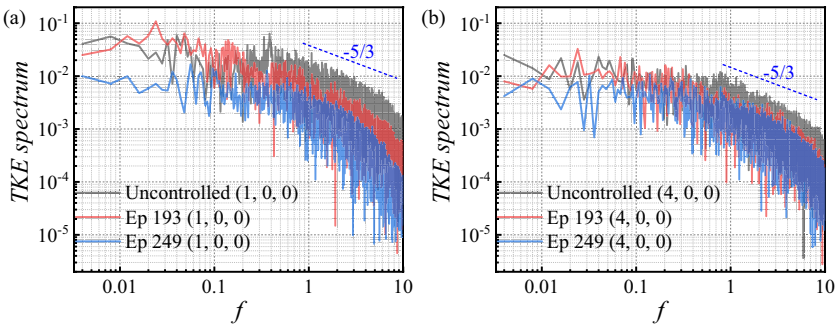


FIG. 11. TKE spectra of the uncontrolled case and the DRL-controlled case learned at the 193rd and 249th episode: (a) monitor point located at (1, 0, 0); (b) monitor point located at (4, 0, 0). Kolmogorov's $-5/3$ scaling law is drawn for reference.

exhibit reduced TKE across most spectral components. Notably, Episode 249 shows lower TKE than both the uncontrolled case and Episode 193 at nearly all frequencies, and this trend is consistent at both monitoring locations. These results demonstrate significant turbulence suppression across virtually all scales under the DRL-guided AFC, with Episode 249 delivering superior performance. This conclusion further supports the observed fluctuation suppression in Figs. 10(a5)–10(c5) and 10(a6)–10(c6).

C. Effect of smoothed action

A prevailing hypothesis suggests that effective DRL in turbulent flows requires high-frequency actuation, i.e., rapid modulation of active forcing [57]. To test this premise, we conduct additional experiments where actions are "smoothed" to attenuate high-frequency components while preserving low-frequency features. In these tests, real-time actions generated by the DRL agent undergo subsequent processing via sequential smoothing. We implement an online second-order exponential smoothing [58,59] for temporal action filtering, which is expressed as

$$\hat{F}^t = \alpha F^t + (1-\alpha)(\hat{F}^{t-1} + b^{t-1}), \quad t = 3, 4, \ldots, \quad b^t = \beta(\hat{F}^t - \hat{F}^{t-1}) + (1-\beta)b^{t-1}, \tag{2}$$

where $\alpha = 1-(1-\delta)^2$ and $\beta = \delta^2/\alpha$. Here F^t denotes the real-time actuation determined by the DRL agent, which is also denoted as an unsmoothed signal in the following results. $\hat{F}^t$ is the smoothed actuation, which is applied as the AFC's actuation. Omitting the term b^{t-1} would lead to a first-order smoothing scheme, i.e., the exponential decaying scheme. Initial conditions are $\hat{F}^1 = F^1$ with $\alpha = \beta = 1$ at $t = 0$. A δ value approaching 1 preserves most original signal content, while δ approaching 0 denoises toward a constant value. Varying δ enables adjustable low-pass filtering intensity for testing the aforementioned assumptions. We select three parameter values, i.e., $\delta = 0.1$, 0.025, and 0.01, which represent a weak, moderate, and strong smoothing operation, respectively. The moderate $\delta = 0.025$ aligns closely with the $\delta = 0.03$ recommended by Ghaffari *et al.* [59].

Figures 12(a1)–12(c1) display time histories of Ω over the initial stage $t \in [0, 10]$, comparing DRL-determined actions (denoted as an "unsmoothed" signal) with their smoothed counterparts. This small time window is chosen for the sake of a clearer view of the differences before and after the smoothing operation. With an identical flow condition at $t = 0$ when the AFC is turned on, the unsmoothed curves are nearly identical at the very beginning, particularly in Figs. 12(b1) and 12(c1), where δ is relatively small. As the actuation and smoothing operation proceed, even the unsmoothed curves begin to become different, because of the online nature of the smoothing operation. Applying different smoothing levels δ to the agent's actions alters the resulting flow fields, which, in turn, affects the subsequent actuation. This is why both the unsmoothed and smoothed curves of Ω look different in the subplots with different δ. For $\delta = 0.1$, smoothing minimally alters the actions while

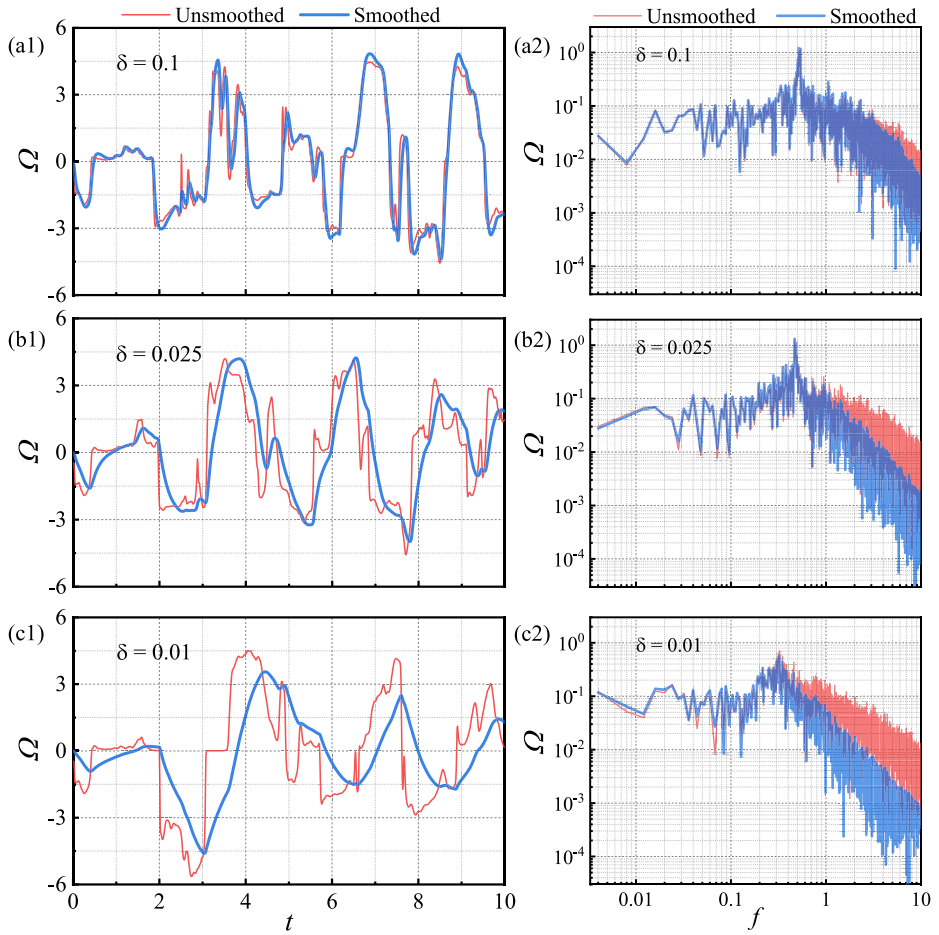


FIG. 12. Time histories and FFT results of unsmoothed and smoothed Ω: (a1)–(a2) $\delta = 0.1$, (b1)–(b2) $\delta = 0.025$, and (c1)–(c2) $\delta = 0.01$.

preserving essential trend features. As δ decreases, the smoothed curves exhibit increased temporal smoothness and greater deviation from the unsmoothed varying trend. Moreover, the amplitude of action is attenuated as evidenced by reduced peak values, and the phase delays appear with temporal shifts.

The FFT results in Figs. 12(a2)–12(c2) compare unsmoothed and smoothed actions in the frequency domain. For all three δ values, low-frequency features remain nearly unchanged while high-frequency components are significantly suppressed. As δ decreases, a wider frequency band is affected. The peak frequency is retained for $\delta = 0.1$ and 0.025 but lost for $\delta = 0.01$.

Control effects using the three smoothing levels are shown in Fig. 13 via direct comparison of the mean drag, drag fluctuations, and lift fluctuations. Data are collected over the interval $t = 50$–300 during the equilibrium state. For clarity, the smoothed results correspond to the strategy learned at the 249th episode without the action smoothing operation, identical to the case Ep 249 presented in Sec. III B. In Fig. 13, drag and lift results at $\delta = 0.1$ remain unaffected. As δ decreases, the mean drag, drag fluctuations, and lift fluctuations all increase. Consequently, control performance degrades when smoothing strength exceeds a certain threshold. However, note that even at $\delta = 0.025$, the reduced mean drag by 49% and the mitigated lift fluctuation by 11%, compared to the uncontrolled case, suggest that the DRL-guided AFC is still very effective.

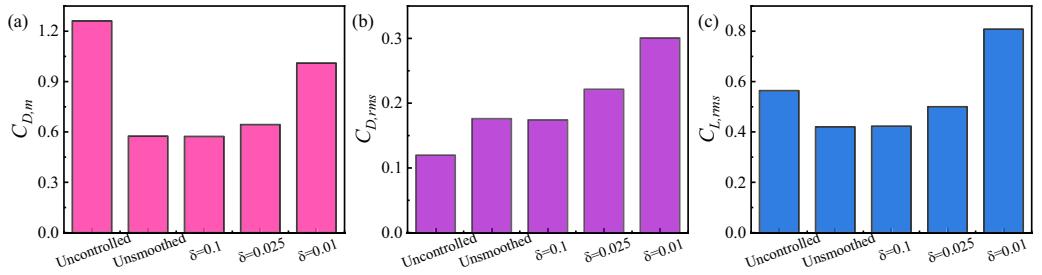


FIG. 13. Comparisons of the drag and lift characteristics due to the effect of smoothed actions. (a) Mean drag, (b) drag fluctuations, and (c) lift fluctuations. The unsmoothed case represents the DRL-guided AFC case learned at the 249th episode.

This section demonstrates notable control effects when applying smoothed actions to AFC. We can now conclusively address the question posed earlier: successful turbulent flow control does not necessarily require high-frequency actions. The critical frequency is still in the same order of frequencies related to vortex shedding and/or shear-layer instability as revealed in Fig. 8. In our view, the high-frequency actions observed in prior studies may primarily stem from the turbulent nature of the flows themselves, which inherently exhibit broadband frequency components, rather than being essential for effective control. In addition, we think that these findings are not universal. The performance impact of action smoothing, whether improvement or degradation, depends critically on both the specific controlled system and its particular control objectives.

IV. SUMMARY

In the present work, we implement DRL-guided AFC for flow past a circular cylinder under 3D and fully developed turbulent conditions at $\mathrm{Re} = 10\,000$, targeting at drag reduction and lift fluctuation mitigation. The key findings are summarized as follows:

(i) An efficient numerical framework that combines the 3D GICLBM flow solver with the PPO algorithm is established. This framework enables ultrafast training, facilitating effective exploration of control strategies in complex 3D turbulent flows.

(ii) A two-stage training strategy is employed: Stage I utilizes a coarse mesh with minimal spanwise resolution to efficiently converge toward an optimal control strategy; Stage II refines the strategy through fine-tuning exploration with a finer mesh and adequate 3D flow configuration. This partitioned training ensures computational efficiency while preserving flow physics essential to the control objectives.

(iii) Analysis of AFC results demonstrates that DRL-guided control successfully achieves pressure recovery, attenuated wake vortical structures, reduced recirculation bubble length, and mitigated downstream turbulent fluctuations. These effects yield a drag reduction of 55% and a lift fluctuation reduction of 26%.

(iv) Additional numerical experiments confirm that effective turbulent flow control does not require high-frequency actuation. This capability is realized through second-order smoothing operations applied to DRL-determined control actions.

This work further establishes DRL as a powerful machine-learning tool for addressing AFC challenges in complex flow systems. Although the geometrical shape and Reynolds number are kept identical in this work, the developed numerical framework and two-stage training strategy are not limited to these conditions and can enable future applications to more intricate flow scenarios such as flow past a rotor sail and flow past an underwater vehicle or its appendages. Moreover, the revealed phenomena and underlying physics suggest rich opportunities for exploring DRL-guided AFC, promising new insights across fluid mechanics and related disciplines.

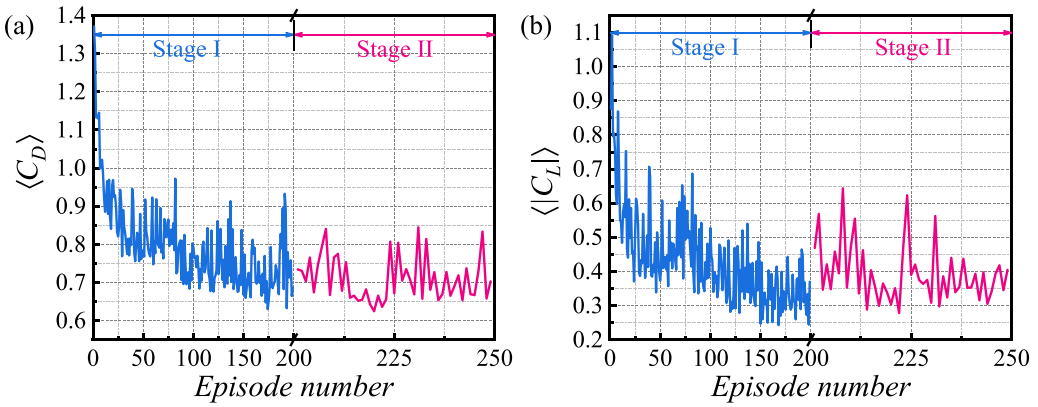


FIG. 14. A supplementary DRL training to showcase the variations of (a) mean drag coefficient $\langle C_D \rangle$ and (b) mean lift fluctuation magnitude $\langle |C_L| \rangle$ during the two-stage training.

ACKNOWLEDGMENTS

We gratefully acknowledge financial support from National Natural Science Foundation of China (52571372, 524713672), the Open Fund of National Key Laboratory on Ship Vibration and Noise (6142204240107), Shenzhen Science and Technology Program (JCYJ20250604183357075), Guangdong Basic and Applied Basic Research Foundation (2026A1515011604), and National Key Research and Development Program of China (2022YFC2805200). The first author also acknowledges financial support from China Scholarship Council during his visit in National University of Singapore.

DATA AVAILABILITY

The data that support the findings of this article are openly available [60].

APPENDIX: SUPPLEMENTARY DRL TRAINING

Due to the stochastic nature of fully developed turbulent flow and the randomness inherent in the DRL exploration, some variation between independent trials can be expected. To provide a straightforward comparison, an additional training trial is performed using identical setups. The overall trends, presented in Fig. 14, align closely with those in Fig. 4, thereby confirming the robustness of the training characteristics observed, even under fully developed turbulent conditions. These comparative findings confirm that multiple trials conducted under identical conditions yield very similar convergence trends, reflecting a relatively stable and consistent underlying learning process.

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